

Notes: Horenstein, Vasquez, and Xiao (RFS, 2026)

Common Factors in Equity Option Returns

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1 Big picture

The paper asks a simple but deep question:

*What is the common factor structure in the cross-section of **delta-hedged equity option returns**, and can we summarize it with a **small set of interpretable, tradable factors**?*

The empirical contribution is two-layered and internally consistent:

1. **Latent factor structure:** Using a large panel of delta-hedged option portfolio returns, the authors estimate how many statistical factors exist (PCA, factor-number tests) and whether calls and puts share the same factors.
2. **Economic factor structure:** They then search for a parsimonious *candidate-factor model* that matches the latent model's ability to price the cross-section. The key result is that a **four-factor** tradable model (EWOP, HV-IV, Cash, VOV) achieves pricing performance comparable to a four-PCA-factor benchmark.

A conceptual interpretation ties everything together:

- Delta-hedged option returns are primarily compensation for **volatility-related risks** (variance risk and volatility-of-volatility risk), plus **option-market-wide** (contract-specific) variation.
- The proposed tradable factors are intended to proxy these risks:
 - (a) HV-IV $\Rightarrow$ volatility risk premium proxy,
 - (b) VOV $\Rightarrow$ volatility-of-volatility risk proxy,
 - (c) Cash $\Rightarrow$ an option-liquidity / balance-sheet proxy,
 - (d) EWOP $\Rightarrow$ an option-market aggregate (contract-specific) factor that is critical for time-series variation.

2 Data and construction of delta-hedged option returns

2.1 Why delta-hedge?

Raw option returns are mechanically sensitive to the underlying stock return. The paper follows the standard approach in the options literature and studies **delta-hedged** option returns so that, locally, the portfolio is not exposed to contemporaneous stock price movements.

2.2 Delta-hedged option gain: continuous-time definition

Let O_t denote the price of a European option written on stock S_t . Let $\Delta_t = \partial O_t / \partial S_t$ denote the option's delta and let r_t denote the annualized risk-free rate. Following Bakshi and Kapadia (2003), the (excess) delta-hedged *gain* over $[t, t + \tau]$ is

$$G_{t,t+\tau} = O_{t+\tau} - O_t - \int_t^{t+\tau} \Delta_u dS_u - \int_t^{t+\tau} r_u (O_u - \Delta_u S_u) du. \quad (1)$$

Intuitively:

- $O_{t+\tau} - O_t$ is the option price change;
- $-\int \Delta_u dS_u$ subtracts the P&L from dynamically hedging with Δ shares of the stock;
- $-\int r_u (O_u - \Delta_u S_u) du$ subtracts the financing return on the (time-varying) cost of the delta-hedged position.

2.3 Delta-hedged option gain: discrete implementation (paper's Equation (1))

In the data, hedging is discrete. Rebalance at dates t_n for $n = 0, 1, \dots, N - 1$ over the holding period. With α_n calendar days between t_n and t_{n+1} , the paper uses

$$G_{t,t+\tau} = O_{t+\tau} - O_t - \sum_{n=0}^{N-1} \Delta_{t_n} (S_{t_{n+1}} - S_{t_n}) - \sum_{n=0}^{N-1} \frac{\alpha_n r_{t_n}}{365} (O_{t_n} - \Delta_{t_n} S_{t_n}). \quad (2)$$

2.4 From gains to comparable returns

A practical issue: delta-hedged gains must be scaled to compare across firms and across calls vs. puts. The paper scales delta-hedged gains by the **stock price** S (following Bakshi–Kapadia and Büchner–Kelly), rather than scaling by the initial net investment $|O - \Delta S|$, because the latter can make put “returns” mechanically more negative than call “returns.” The stock-price scaling produces call/put comparability.

2.5 Sample and portfolio formation in words

Key data inputs and filters (as described in the paper):

- Option data from **OptionMetrics**;
- Monthly portfolio formation; hold from end of month t to end of month $t + 1$;
- Daily delta hedging; when delta is missing, use the previous day's delta (robustness with interpolation);

- Retain options with economically meaningful quotes/volume and enforce basic no-arbitrage filters;
- Focus on options with moneyness in $[0.8, 1.2]$; choose the option closest to at-the-money with the shortest maturity among those with more than one month to maturity.

3 Test assets and candidate factors

3.1 ATM test portfolios

The main test assets are **ATM delta-hedged option portfolio returns** formed by sorting options on predictors. Concretely:

- 185 ATM **call** portfolios and 185 ATM **put** portfolios;
- portfolios are **deciles** (10 bins) for most characteristics; **quintiles** for credit rating;
- the combined panel therefore has $N = 370$ portfolios.

3.2 The 19 characteristics used to form portfolios (core candidate list)

At month-end, sort firms by each characteristic and compute the subsequent-month delta-hedged option returns:

- (1) Size: log market equity.
- (2) Ivol: idiosyncratic volatility of the underlying stock.
- (3) HV–IV: log difference between historical realized volatility and Black–Scholes implied volatility (ATM).
- (4) IV_slope: slope of the volatility term structure (long minus short IV).
- (5) BM: book-to-market ratio.
- (6) Credit: S&P credit rating mapped to numeric values.
- (7) VOV: volatility of implied volatility (past-month IV change dispersion).
- (8) Illiquidity: Amihud (2002) illiquidity.
- (9) Reversal: 1-month lagged stock return.
- (10) Mom: stock momentum (11 months ending at beginning of previous month).
- (11) VarCF: variance of cash-flow-to-market-equity over 60 months.

- (12) Cash: cash holdings over total assets.
- (13) Disp: analyst earnings forecast dispersion.
- (14) ShareIss1Y: 1-year change in shares outstanding.
- (15) ShareIss5Y: 5-year change in shares outstanding.
- (16) PM: profit margin (EBIT / revenues).
- (17) Close: log stock price.
- (18) Profit: profitability (earnings / book equity).
- (19) XFIN: total external financing (net equity + net debt issuance minus dividends), scaled by assets.

3.3 Candidate factor construction

For each characteristic x , define the (ATM) **long–short** option factor as

$$F_t^x \equiv R_t^{x,\text{high}} - R_t^{x,\text{low}},$$

where “high” is decile 10 (or quintile 5 for credit) and “low” is decile 1 (or quintile 1).

In addition to the 19 characteristic-based long–short factors, the paper adds two contract-market candidates:

- (a) **SPX_DH_call / SPX_DH_put**: delta-hedged S&P 500 index option return (a market-volatility-risk proxy used in earlier literature).
- (b) **EWOP**: the **equal-weighted option portfolio return** of the full set of characteristic-sorted portfolios, intended to capture aggregate option-market variation.

3.4 Interpreting Table 1 (summary statistics of candidate factors)

Table 1 summarizes the distribution of the 21 candidate factor returns for calls and puts (means, volatility, percentiles, skewness, kurtosis).

My interpretation:

- Candidate factor means vary substantially in sign and magnitude, which already suggests a nontrivial option-return cross-section.
- The most prominent average spreads (in absolute value) are associated with **HV–IV** and **VOV**, foreshadowing their later role as core factors.

- The index delta-hedged factor has extreme non-normality (very high kurtosis), indicating that it may be a noisy “market” proxy once we move from index options to individual-stock options.

4 Empirical factor model setup

4.1 Approximate factor model (page 12)

Let r_{it} be the return on delta-hedged option portfolio i at time t . The paper models the $N \times 1$ return vector as

$$\mathbf{r}_t = \boldsymbol{\alpha} + \mathbf{B}\mathbf{f}_t + \boldsymbol{\varepsilon}_t, \quad (3)$$

where:

- $\mathbf{f}_t \in \mathbb{R}^K$ is a vector of K common factors;
- $\mathbf{B}$ is the $N \times K$ matrix of factor loadings;
- $\boldsymbol{\varepsilon}_t$ are idiosyncratic components allowed to be weakly correlated across i and t .

4.2 How they evaluate a factor model (objects used throughout)

For a given factor specification, the paper uses a consistent set of diagnostics:

- (a) **Time-series fit:** For each test asset i , run

$$r_{i,t} = \alpha_i + \beta'_i \mathbf{f}_t + u_{i,t}.$$

Report the *average adjusted* R^2 across assets.

- (b) **Cross-sectional pricing fit:** Compute predicted average returns $\hat{\beta}'_i \bar{\mathbf{f}}$ (with full-sample $\hat{\beta}_i$ and sample mean factor returns $\bar{\mathbf{f}}$), and report

$$\text{Corr}(\bar{r}_i, \hat{\beta}'_i \bar{\mathbf{f}}),$$

i.e., the correlation between average and predicted returns across portfolios.

- (c) **Pricing-error magnitude:** Define in-sample RMSPE as

$$\text{RMSPE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \hat{\alpha}_i^2}.$$

Also compute the fraction of portfolios with large intercepts: $\%(|t_{\hat{\alpha}}| > 3)$.

- (d) **Out-of-sample RMSPE:** Using a rolling window (120 months in the paper), estimate factor loadings/scores in window $t - 119, \dots, t$ and evaluate pricing errors in $t + 1$.
- (e) **SDF / mean-variance efficiency test:** The paper uses the distribution-free $F_{\max}$ test of Gungor and Luger (2016), implemented via Monte Carlo resampling, to evaluate whether pricing errors are jointly consistent with mean-variance efficiency.

4.3 Sharpe ratio for delta-hedged factor portfolios (SRDH)

The paper reports the Sharpe ratio of the **tangency portfolio constructed from the factor returns**, but emphasizes that delta-hedged option portfolios require dynamic hedging in the underlying stock, so the resulting Sharpe ratio is not directly comparable to stock/bond Sharpe ratios. The notation SRDH is used to emphasize this.

If $\mathbf{f}_t$ are factor returns with mean $\boldsymbol{\mu}_f$ and covariance $\boldsymbol{\Sigma}_f$, the tangency weights (up to scale) satisfy

$$\mathbf{w}^* \propto \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f,$$

and the (monthly) tangency Sharpe ratio is

$$\text{SRDH} = \frac{\mathbf{w}^{*'} \boldsymbol{\mu}_f}{\sqrt{\mathbf{w}^{*'} \boldsymbol{\Sigma}_f \mathbf{w}^*}}.$$

5 How many factors are there? Calls vs puts? (Figure 1 and Table 2)

5.1 Figure 1: Scree plot and factor-number tests

Figure 1 provides a “Scree” plot of the leading eigenvalues of the sample second-moment matrix of doubly demeaned portfolio returns, along with the Ahn–Horenstein (2013) eigenvalue-ratio and growth-ratio estimators.

Interpretation:

- The leading eigenvalues exhibit a steep drop after roughly three components, indicating **about three strong latent factors**.
- The Ahn–Horenstein diagnostics reinforce the “three strong factors” conclusion, while leaving open the possibility of additional weaker factors.

5.2 Table 2: residual-factor test and contract-specific factors

Table 2 applies the Gagliardini–Ossola–Scaillet (2019) residual-factor test. The logic is:

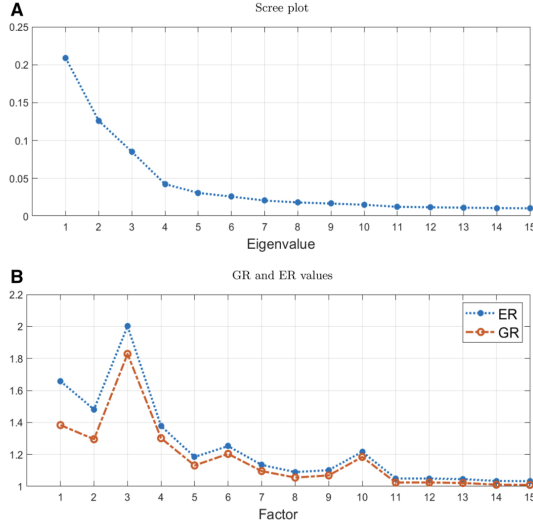


Figure 1
Scree test for at-the-money call and put option returns

Table 1
Summary statistics of the candidate factors in the option market

	Mean	SD	10th. pctl.	25th. pctl.	50th. pctl.	75th. pctl.	90th. pctl.	Skew	Kurt
A: Call options									
Size	0.45	0.76	-0.43	-0.02	0.42	0.85	1.33	0.82	1.86
Ivol	-0.45	0.82	-1.30	-0.88	-0.50	-0.03	0.36	0.61	3.44
HV-IV	0.78	0.78	-0.06	0.27	0.69	1.14	1.73	0.88	1.40
IV_slope	0.51	0.73	-0.31	0.15	0.48	0.91	1.37	0.16	1.99
BM	0.16	0.59	-0.54	-0.10	0.19	0.48	0.83	-0.66	3.44
Credit	-0.18	0.51	-0.80	-0.51	-0.17	0.06	0.39	0.51	2.03
VOV	-0.59	0.78	-1.39	-1.00	-0.58	-0.17	0.23	-0.45	2.94
Illiquidity	-0.43	0.73	-1.21	-0.81	-0.39	0.05	0.37	-0.70	1.95
Reversal	0.04	0.73	-0.73	-0.32	0.06	0.50	0.78	-0.75	3.23
Mom	-0.01	0.73	-0.80	-0.44	-0.05	0.36	0.86	0.01	2.36
VarCF	-0.30	0.60	-1.03	-0.61	-0.25	0.05	0.32	-0.64	2.16
Cash	-0.45	0.77	-1.28	-0.89	-0.48	-0.02	0.46	0.04	2.66
Disp	-0.16	0.55	-0.82	-0.49	-0.15	0.15	0.49	0.19	1.94
Sharelss1y	-0.13	0.57	-0.76	-0.45	-0.13	0.15	0.52	0.61	4.16
Sharelss5y	0.03	0.49	-0.47	-0.23	-0.03	0.25	0.60	1.13	3.42
PM	0.40	0.64	-0.33	0.05	0.40	0.77	1.08	0.54	2.66
Close	0.50	0.77	-0.34	0.04	0.47	0.89	1.37	0.71	2.78
Profit	0.17	0.58	-0.54	-0.10	0.21	0.51	0.80	-0.81	2.64
XFIN	-0.30	0.68	-1.07	-0.66	-0.34	0.05	0.40	0.51	5.30
SPX_DH_Call	-0.06	0.60	-0.63	-0.36	-0.12	0.16	0.51	3.02	22.80
EWOP	-0.04	0.73	-0.74	-0.45	-0.16	0.23	0.78	2.12	10.30
Mean	SD	10th. pctl.	25th. pctl.	50th. pctl.	75th. pctl.	90th. pctl.	Skew	Kurt	
B: Put options									
Size	0.33	0.70	-0.51	-0.06	0.29	0.73	1.07	0.48	1.13
Ivol	-0.40	0.82	-1.19	-0.88	-0.48	0.01	0.50	0.53	3.26
HV-IV	0.80	0.77	-0.04	0.28	0.71	1.18	1.83	0.94	1.79
IV_slope	0.55	0.77	-0.24	0.15	0.50	0.93	1.44	0.39	2.66
BM	0.09	0.58	-0.63	-0.18	0.16	0.41	0.72	-0.66	3.87
Credit	-0.03	0.49	-0.55	-0.32	-0.04	0.19	0.49	0.72	2.99
VOV	-0.56	0.76	-1.33	-0.96	-0.55	-0.13	0.28	-0.67	3.52
Illiquidity	-0.32	0.70	-1.11	-0.68	-0.29	0.09	0.49	-0.63	1.62
Reversal	-0.09	0.72	-0.81	-0.46	-0.03	0.36	0.64	-1.07	4.53
Mom	0.05	0.72	-0.72	-0.37	0.04	0.48	0.86	-0.07	1.89
VarCF	-0.30	0.60	-0.94	-0.58	-0.24	0.05	0.31	-0.28	3.80
Cash	-0.31	0.70	-1.10	-0.69	-0.35	0.06	0.51	0.42	2.28
Disp	-0.16	0.55	-0.78	-0.49	-0.19	0.11	0.45	0.78	3.27
Sharelss1y	-0.04	0.56	-0.67	-0.39	-0.10	0.21	0.58	1.26	4.03
Sharelss5y	0.14	0.49	-0.36	-0.17	0.06	0.34	0.76	1.05	2.72
PM	0.36	0.62	-0.49	-0.01	0.37	0.72	1.04	0.15	0.87
Close	0.41	0.70	-0.45	0.07	0.46	0.71	1.17	0.12	2.27
Profit	0.09	0.59	-0.62	-0.17	0.13	0.43	0.79	-0.89	2.70
XFIN	-0.24	0.67	-0.94	-0.58	-0.31	0.08	0.54	0.73	4.40
SPX_DH_Put	-0.41	1.20	-1.49	-1.01	-0.56	0.06	0.76	3.39	28.45
EWOP	-0.12	0.74	-0.76	-0.53	-0.24	0.11	0.73	2.29	11.65

Estimate common factors from the combined call+put panel, add them sequentially, and ask whether the remaining residuals in the call panel (or in the put panel) still contain common factors.

Interpretation:

- As more combined-panel factors are included, the number of factors detected in the residuals declines.
- The joint pattern is consistent with **mostly shared factors across calls and puts**, but with **contract-specific components** (i.e., factors that are not fully shared across the two contract types).

6 Latent factors via PCA (Figure 2, Tables 3–4)

6.1 PCA extraction and the benchmark “latent” factor model

The paper uses PCA to estimate latent factors and loadings from the panel of option portfolio returns. Practically:

- Construct the return panel for $N = 370$ ATM portfolios over $T = 312$ months (1996–2021).

Table 2
Estimation for the number of factors in the delta-hedged call and put option portfolios

Common factors used	Number of factors found in the residuals	
	Call options	Put options
1	3	2
2	2	2
3	1	1
4	0	0

This table presents results based on the Gagliardini, Ossola, and Scaillet (GOS) estimator proposed by Gagliardini, Ossola, and Scaillet (2019). The GOS estimator scrutinizes the error terms generated by a factor model and tests whether these errors are weakly cross-sectionally correlated or share at least one common factor. This table shows the number of factors found in the residuals of the call and put data sets as we sequentially add one of the common factors estimated from the combined data sets. The factors are simultaneously estimated from the combined data sets of calls and puts. The test assets are 370 option portfolios: 185 delta-hedged ATM call and 185 delta-hedged ATM put option portfolios sorted on the following 19 characteristics: firm size, idiosyncratic volatility, HV-IV, the slope of volatility term structure, book to market, S&P credit rating, VOV, Amihud (2002) illiquidity measure, return reversal, return momentum, cash flow variance, cash-to-assets ratio, analyst earnings forecast dispersion, 1-year and 5-year new issues, profit margin, stock price, profitability, and total external financing. Sample period is from January 1996 to December 2021 for stocks in the OptionMetrics database.

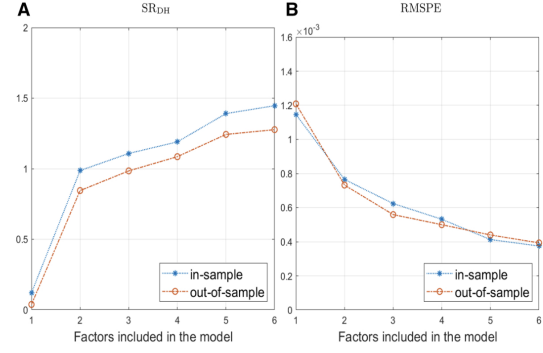


Figure 2
In-sample and out-of-sample performance of the PCA factors

- Demean (“double demean”) appropriately and compute leading principal components.
- Evaluate models with $K = 1, \dots, 6$ PCA factors.

6.2 Figure 2: in-sample vs out-of-sample performance as K grows

Figure 2 plots SRDH and RMSPE as we add PCA factors.

Interpretation:

- Adding factors increases the achievable tangency Sharpe ratio and reduces pricing errors.
- Performance gains are largest for the first few factors and then exhibit diminishing returns.
- The out-of-sample curves are crucial: the factor structure is not a pure in-sample artifact.

6.3 Table 3: PCA factor model performance

Table 3 reports full metrics for PCA models with 1–6 factors, for:

- Call+put portfolios together (panel A),
- Call portfolios only (panel B),
- Put portfolios only (panel C).

How to read the results:

- **Cross-sectional fit becomes very high by 4 factors.** In panel A, $\text{Corr}[\bar{r}, \hat{r}]$ reaches about 0.90 with four PCA factors, and average adjusted R^2 is above 90%.

- **Calls and puts look similar.** Panels B and C show that calls-only and puts-only panels exhibit comparably strong factor structure.
- **Mean-variance efficiency is more demanding.** The $F_{\max}$ test rejects low- K models; higher- K models can yield inconclusive/less-rejecting outcomes. This motivates using PCA as a *benchmark* and then seeking an interpretable model that matches its pricing performance.

Table 3
Performance of PCA factor models with different number of factors

Models	SR _{DH}	Corr Avg(r)-Pred(r)	Average adj R^2 (%)	RMSPE (%)	$ t_a > 3$ (%)	OOS RMSPE (%)	$F_{\max}$
A: PCA factors from 370 call and put portfolios							
Factor 1	0.12	0.59	86	0.11	52	0.12	0.00
Factor 2	0.99	0.80	88	0.08	30	0.07	0.00
Factor 3	1.11	0.87	90	0.06	21	0.06	0.00
Factor 4	1.19	0.90	91	0.05	20	0.05	0.00
Factor 5	1.39	0.94	91	0.04	9	0.04	0.00,0.65
Factor 6	1.45	0.95	92	0.04	8	0.04	0.00,0.63
B: PCA factors from 185 call portfolios							
Factor 1	0.07	0.68	87	0.11	62	0.11	0.00
Factor 2	0.89	0.89	89	0.05	14	0.05	0.00
Factor 3	0.92	0.91	91	0.05	18	0.05	0.00
Factor 4	1.17	0.95	91	0.03	8	0.04	0.00,0.72
Factor 5	1.22	0.96	92	0.03	6	0.04	0.00,0.99
Factor 6	1.24	0.97	92	0.03	5	0.03	0.00,1
C: PCA factors from 185 put portfolios							
Factor 1	0.17	0.53	88	0.10	49	0.11	0.00
Factor 2	0.82	0.80	90	0.07	34	0.05	0.00
Factor 3	1.03	0.88	91	0.05	26	0.05	0.00
Factor 4	1.29	0.95	92	0.03	4	0.04	0.00,0.73
Factor 5	1.31	0.96	92	0.03	4	0.03	0.00,1
Factor 6	1.32	0.96	92	0.03	4	0.03	0.00,1

Table 4
Performance of PCA factor models in the orthogonalized portfolios

Models	SR _{DH}	Corr Avg(r)-Pred(r)	Average adj R^2 (%)	RMSPE (%)	$ t_a > 3$ (%)	OOS RMSPE (%)	$F_{\max}$
A: PCA factors from 185 orthogonalized call and 185 orthogonalized put portfolios							
Factor 1	0.90	0.83	13	0.06	24	0.05	0.00
Factor 2	1.00	0.88	21	0.05	25	0.05	0.00
Factor 3	1.28	0.94	24	0.04	7	0.04	0.00,0.22
Factor 4	1.33	0.95	26	0.03	5	0.04	0.00,0.30
Factor 5	1.35	0.95	29	0.03	6	0.04	0.00,0.59
Factor 6	1.36	0.96	30	0.03	5	0.03	0.00,0.95
B: PCA factors from 185 orthogonalized call portfolios							
Factor 1	0.89	0.83	12	0.06	23	0.05	0.00
Factor 2	0.92	0.86	20	0.06	26	0.05	0.00
Factor 3	1.17	0.94	23	0.04	9	0.04	0.00,0.48
Factor 4	1.22	0.95	25	0.03	8	0.04	0.00,0.98
Factor 5	1.24	0.95	27	0.03	7	0.03	0.00,0.98
Factor 6	1.24	0.95	29	0.03	7	0.03	0.00,1
C: PCA factors from 185 orthogonalized put portfolios							
Factor 1	0.81	0.82	12	0.06	23	0.06	0.00
Factor 2	1.02	0.86	19	0.05	24	0.05	0.00
Factor 3	1.28	0.91	22	0.04	11	0.04	0.00,0.45
Factor 4	1.30	0.92	25	0.04	10	0.03	0.00,0.98
Factor 5	1.31	0.92	27	0.04	10	0.03	0.01,1
Factor 6	1.31	0.92	29	0.04	11	0.03	0.00,1

6.4 Table 4: identifying the contract-specific PCA factor

A key identification step is: which PCA factor is contract-specific?

The paper orthogonalizes call and put returns with respect to each of the first four PCA factors (separately), recombines orthogonalized calls and puts, and re-runs PCA. The logic is:

If I remove the contract-specific factor from each contract type, the remaining factors should be common across calls and puts, so a **three-factor** PCA should suffice on the combined orthogonalized data.

Interpretation:

- Only when orthogonalizing with respect to the **first** PCA factor does a three-factor PCA become sufficient for the combined data.
- Therefore, the **first** PCA factor in calls and the **first** PCA factor in puts are the **contract-specific factors**.
- Combining this with the earlier factor-number evidence yields a clean structure:

Calls: 3 common factors+1 call-specific factor, Puts: 3 common factors+1 put-specific factor.

Hence, the combined panel needs $3 + 2 = 5$ factors, while each contract type alone needs $3 + 1 = 4$.

7 From latent factors to tradable economic factors (Figures 3–5, Tables 5–6)

7.1 Step 1: candidate-factor correlation structure (Figure 3)

Figure 3 shows an absolute correlation heatmap for the 19 characteristic-based call factors and the 19 characteristic-based put factors (plus EWOP).

Interpretation:

- For the same characteristic, call and put long–short factors are highly correlated in absolute value (later quantified around ≈ 0.85).
- This supports the paper’s central message: calls and puts share a largely common factor structure in their delta-hedged returns.

7.2 Step 2: searching for a parsimonious 4-factor candidate model (Figure 4)

The paper constructs 21 candidate factors (19 characteristics + EWOP + SPX delta-hedged index option factor) and generates *all* possible 4-factor models:

$$\binom{21}{4} = 5,985.$$

For each model, it computes five performance metrics (SRDH; ability to span the PCA tangency portfolio; corr between average and predicted returns; RMSPE; OOS RMSPE), selects the top 2.5% models for each metric, and records how often each candidate factor appears among the top models.

Figure 4 summarizes the frequencies. My interpretation:

- **HV–IV and Cash** are the dominant cross-sectional factors: they appear in almost all top-performing models for multiple criteria.
- **VOV** is the next most important factor, improving fit beyond HV–IV and Cash.
- **EWOP** is essential as the option-market aggregate/time-series factor (closely related to the first PCA factor).
- The **SPX delta-hedged** factor rarely appears among the top models, suggesting it is not a clean market aggregator for individual-stock option returns.

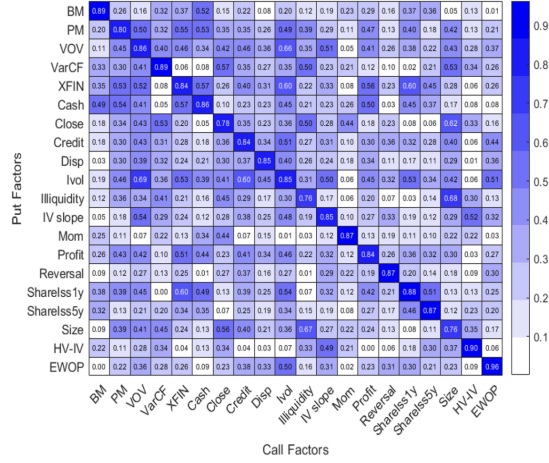


Figure 3
Correlation heatmap of at-the-money call and put factors

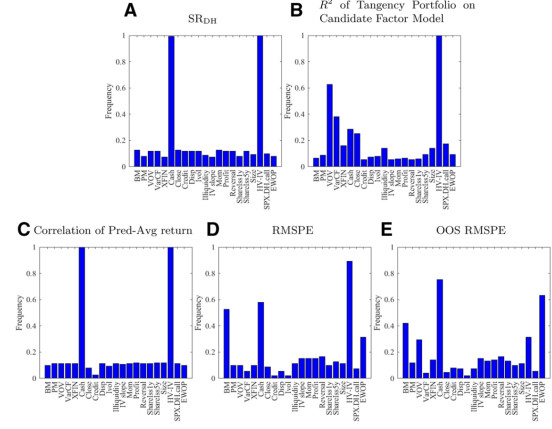


Figure 4
Candidate factor selection

7.3 Step 3: pricing performance of the chosen tradable model (Table 5 and Figure 5)

Table 5 evaluates several tradable specifications (calls and puts), with the key benchmark being the 4-factor model:

$$\{\text{EWOP}, \text{HV-IV}, \text{Cash}, \text{VOV}\}.$$

Table 5
Performance of models with tradable factors

Models	SR _{OH}	Corr Avg(r)-Pred(r)	Average adj R ² (%)	RMSPE (%)	t ₁ > 3 (%)	OOS RMSPE (%)	F _{max}
A: Call candidate factors on call portfolio							
EWOP, HV-IV, Cash	1.20	0.96	89	0.033	4	0.050	0.04.1
EWOP, HV-IV, Cash, VOV	1.22	0.96	90	0.034	4	0.047	0.04.1
EWOP, HV-IV, Cash, BM	1.20	0.96	90	0.034	5	0.050	0.06
EWOP, HV-IV, Cash, VOV, BM	1.22	0.96	90	0.034	4	0.047	0.04.1
EWOP, HV-IV, Cash, VOV, VaICF	1.23	0.96	90	0.034	6	0.047	0.04.1
B: Put candidate factors on put portfolio							
EWOP, HV-IV, Cash	1.18	0.94	89	0.048	10	0.067	0.00.54
EWOP, HV-IV, Cash, VOV	1.21	0.95	89	0.040	6	0.054	0.00.97
EWOP, HV-IV, Cash, BM	1.19	0.94	89	0.048	10	0.066	0.00.78
EWOP, HV-IV, Cash, VOV, BM	1.21	0.95	90	0.040	5	0.053	0.00.1
EWOP, HV-IV, Cash, VOV, VaICF	1.21	0.95	90	0.040	6	0.053	0.00.1
C: Call candidate factors on put portfolio							
EWOP, HV-IV, Cash	1.20	0.92	90	0.045	12	0.059	0.00.83
EWOP, HV-IV, Cash, VOV	1.21	0.91	90	0.044	11	0.053	0.00.98
EWOP, HV-IV, Cash, BM	1.20	0.91	90	0.046	12	0.058	0.00.94
EWOP, HV-IV, Cash, VOV, BM	1.21	0.91	90	0.045	12	0.053	0.00.1
EWOP, HV-IV, Cash, VOV, VaICF	1.21	0.91	90	0.044	13	0.052	0.00.1
D: Put candidate factors on call portfolio							
EWOP, HV-IV, Cash	1.18	0.92	90	0.042	10	0.058	0.00.98
EWOP, HV-IV, Cash, VOV	1.19	0.92	91	0.042	12	0.049	0.00.69
EWOP, HV-IV, Cash, BM	1.18	0.92	90	0.044	12	0.059	0.00.99
EWOP, HV-IV, Cash, VOV, BM	1.20	0.92	91	0.043	13	0.050	0.00.67
EWOP, HV-IV, Cash, VOV, VaICF	1.19	0.93	91	0.042	12	0.049	0.00.83

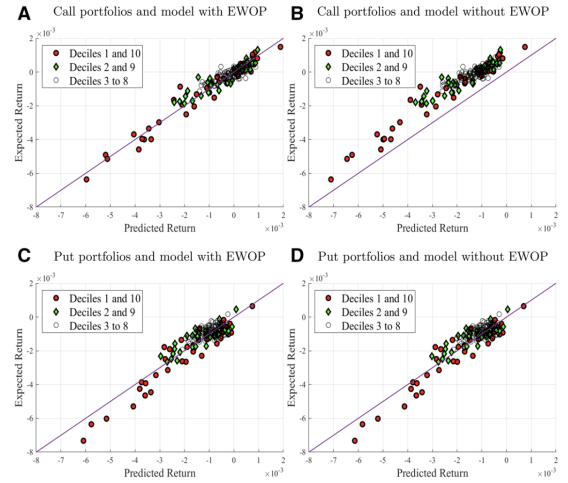


Figure 5
Pricing performance of the candidate factor models

Interpretation of Table 5:

- The parsimonious models achieve **very high** $\text{Corr}[\bar{r}, \hat{r}]$ (around 0.9–0.96) and low RMSPE, comparable to the 4-PCA-factor benchmark.

- The results are robust across whether factors are constructed from call sorts or put sorts and across whether the test assets are call portfolios or put portfolios.
- Adding extra factors like BM or VarCF provides marginal improvements but is not essential for the main result.

Figure 5 visualizes the average-vs-predicted return relationship for:

- 4 PCA factors,
- a large model with all candidate factors,
- smaller candidate models (with/without EWOP; with/without Cash).

Interpretation:

- The selected parsimonious model fits the **extreme deciles** particularly well (which are the economically important portfolios in long–short spreads).
- EWOP matters more for **pricing errors** / **intercepts** than for the slope of the cross-sectional average-vs-predicted relation.

7.4 Table 6: do the 4 factors “explain” the other long–short strategies?

Table 6 reports:

- raw means of each long–short characteristic strategy, and
- its alpha with respect to the 4-factor model (EWOP, HV–IV, Cash, VOV),

for both calls and puts.

Interpretation:

- Many raw long–short returns are statistically significant, confirming a rich option-return cross-section.
- The 4-factor model drives most of these alphas toward zero: the model *spans* much of the characteristic-sorted variation.
- The high correlation between call-based and put-based factors strengthens the view that the core priced risks are shared across contracts.

Table 6
Summary statistics of the long-short factors

	Delta-hedged call options		Delta-hedged put options	
	Mean	4-factor alpha	Mean	4-factor alpha
Size	0.16% (4.90)	0.04% (0.90)	0.09% (2.89)	0.05% (0.99)
Ivol	0.40% (11.12)	0.00% (−0.07)	0.36% (10.16)	0.05% (1.11)
IV_slope	−0.30% (−8.83)	−0.07% (−1.44)	−0.30% (−8.71)	−0.07% (−1.51)
BM	−0.30% (−7.65)	0.01% (0.37)	−0.24% (−6.15)	0.01% (0.15)
Credit	0.50% (11.53)	0.04% (0.71)	0.41% (10.31)	0.09% (1.57)
Illiquidity	−0.18% (−6.35)	−0.07% (−1.87)	−0.03% (−1.19)	0.09% (2.38)
Reversal	−0.16% (−5.11)	−0.04% (−0.95)	−0.16% (−4.98)	−0.07% (−1.47)
Mom	−0.45% (−9.65)	−0.06% (−1.33)	−0.40% (−8.72)	−0.05% (−0.99)
VarCF	−0.43% (−10.34)	−0.01% (−0.14)	−0.32% (−7.95)	0.03% (0.55)
Disp	0.51% (12.40)	−0.01% (−0.22)	0.55% (12.57)	0.02% (0.27)
shareiss1	−0.01% (−0.20)	−0.12% (−2.00)	0.05% (1.34)	0.01% (0.12)
shareiss5	0.17% (5.15)	−0.06% (−1.34)	0.09% (2.78)	−0.12% (−2.66)
PM	0.04% (1.06)	−0.10% (−1.37)	−0.09% (−2.13)	−0.17% (−2.24)
Close	−0.13% (−3.97)	0.03% (0.71)	−0.04% (−1.41)	0.04% (0.83)
Profit	0.03% (1.19)	0.04% (1.07)	0.14% (4.94)	0.11% (2.94)
XFIN	0.45% (10.41)	−0.05% (−0.90)	0.33% (8.44)	−0.04% (−0.71)

Table 7
Pricing performance on out-of-the-money option portfolios

Models	SR _{DH}	Corr Avg(r)-Pred(r)	Average adj R^2 (%)	RMSPE (%)	$ t_\alpha > 3$ (%)	OOS RMSPE (%)	F_{max}
<i>A: ATM call candidate factors on OTM call portfolios</i>							
EWOP, HV-IV, Cash	1.20	0.91	66	0.07	4	0.09	0.01,1
EWOP, HV-IV, Cash, VOV	1.22	0.91	67	0.08	9	0.11	0.00,1
<i>B: ATM put candidate factors on OTM call portfolios</i>							
EWOP, HV-IV, Cash	1.18	0.91	65	0.06	2	0.07	0.19
EWOP, HV-IV, Cash, VOV	1.21	0.92	66	0.07	5	0.10	0.03,1
<i>C: ATM call candidate factors on OTM put portfolios</i>							
EWOP, HV-IV, Cash	1.20	0.88	74	0.05	3	0.05	0.00,0.98
EWOP, HV-IV, Cash, VOV	1.21	0.89	74	0.05	2	0.05	0.00,1
<i>D: ATM put candidate factors on OTM put portfolios</i>							
EWOP, HV-IV, Cash	1.18	0.88	74	0.05	4	0.05	0.00,0.6
EWOP, HV-IV, Cash, VOV	1.19	0.90	74	0.05	3	0.05	0.00,0.95

8 Beyond ATM: OTM options and 150-characteristic robustness (Tables 7–9, Figure 6)

8.1 OTM pricing (Table 7)

The paper constructs an OTM subsample where each firm-month has ATM call, ATM put, OTM call (moneyness near 1.1), and OTM put (moneyness near 0.9). Then it applies the ATM-derived factor model to OTM portfolio returns.

Interpretation of Table 7:

- ATM-based factors price OTM portfolios well: the correlation between average and predicted returns remains high (around ≈ 0.9).
- This suggests that the factor structure is not unique to ATM contracts; it extends across moneyness.
- Including VOV can improve some metrics but the core pricing information is already captured by EWOP, HV–IV, and Cash.

8.2 Robustness with 150 characteristics (Tables 8–9, Figure 6)

To probe robustness, the paper switches to 150 characteristics (from Chen–Zimmermann’s Open Asset Pricing set) and forms:

- 1,495 decile portfolios for calls and 1,495 for puts (2,990 total),
- 150 long–short call portfolios and 150 long–short put portfolios.

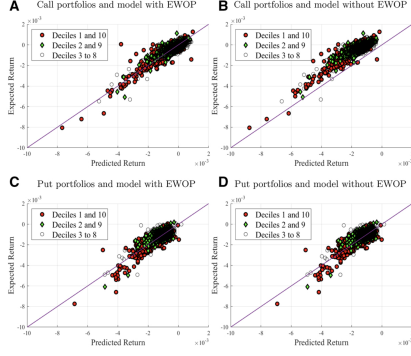


Figure 6
Pricing performance of the candidate factor models: 2,990 portfolios sorted by 150 characteristics

Table 8. Table 8 summarizes the distribution of the *absolute value* of long–short returns (and their t -stats) across the 150-characteristic sorts.

Interpretation:

- Many characteristic-sorted long–short returns are economically nontrivial and statistically strong (a large fraction have $|t| > 3$).
- This reinforces the premise: option returns exhibit a strong factor structure (even stronger than in stock returns, as emphasized in the paper).

Table 9 and Figure 6. Table 9 and Figure 6 show that the parsimonious candidate model (EWOP, HV–IV, Cash, optionally VOV) continues to perform well on this much larger set of test assets.

Interpretation:

- Cross-sectional fit remains very high on the enlarged universe, and pricing errors remain small.
- Visually (Figure 6), the candidate model delivers an average-vs-predicted return relationship close to the PCA benchmark even when the number of portfolios increases dramatically.

9 Appendix theory: why volatility and volatility-of-volatility should matter (Appendix A.1)

The appendix provides a stylized multifactor framework connecting delta-hedged option gains to risk premia for volatility factors and volatility-of-volatility factors.

9.1 Dynamics of stock price and volatility (Equations (A1)–(A4))

For firm i , stock price S_t^i and volatility V_t^i evolve as:

$$\frac{dS_t^i}{S_t^i} = \mu_t^i(S_t^i, V_t^i) dt + V_t^i dW_{1t}^i, \quad (\text{A1})$$

$$V_t^i = \sum_{j=1}^n \beta^j V_{f,t}^j + Z_t^i, \quad (\text{A2})$$

$$dV_{f,t}^j = \theta^j dt + \eta_t^j dW_{2t}^{i,j}, \quad (\text{A3})$$

$$d\eta_t^j = \xi^j dt + q^j dW_{3t}^{i,j}. \quad (\text{A4})$$

Here $V_{f,t}^j$ are common volatility factors and η_t^j governs their volatility (vol-of-vol). Brownian motions are assumed orthogonal for simplicity.

9.2 Option price expansion and no-arbitrage valuation (Equations (A5)–(A6))

Let C_t^i denote the call option price at time t . By Itô's lemma,

$$C_{t+\tau}^i = C_t^i + \int_t^{t+\tau} \Delta_u^i dS_u^i + \int_t^{t+\tau} \sum_{j=1}^n \frac{\partial C^i}{\partial V_f^j} dV_{f,u}^j + \int_t^{t+\tau} \sum_{j=1}^n \frac{\partial C^i}{\partial \eta^j} d\eta_u^j + \int_t^{t+\tau} b_u^i du, \quad (\text{A5})$$

where $\Delta_u^i = \partial C_u^i / \partial S_u^i$ and b_u^i collects second-derivative terms.

No-arbitrage implies a valuation PDE:

$$\begin{aligned} & \frac{1}{2} (V^i S^i)^2 \frac{\partial^2 C^i}{\partial (S^i)^2} + \frac{1}{2} \sum_{j=1}^n (\eta^j)^2 \frac{\partial^2 C^i}{\partial (V_f^j)^2} + \frac{1}{2} \sum_{j=1}^n (q^j)^2 \frac{\partial^2 C^i}{\partial (\eta^j)^2} + r S^i \frac{\partial C^i}{\partial S^i} \\ & + \sum_{j=1}^n (\theta^j - \lambda_v^j) \frac{\partial C^i}{\partial V_f^j} + \sum_{j=1}^n (\xi^j - \lambda_\eta^j) \frac{\partial C^i}{\partial \eta^j} + \frac{\partial C^i}{\partial t} - r C^i = 0. \end{aligned} \quad (\text{A6})$$

The quantities λ_v^j and λ_η^j are risk premia associated with volatility and vol-of-vol factor shocks.

9.3 Delta-hedged gains and expected gains (Equations (A7)–(A9))

Combining (A5) and (A6) yields an expression for the option price change with explicit compensation terms for volatility risks, leading to the delta-hedged gain definition

$$G_{t,t+\tau}^i = C_{t+\tau}^i - C_t^i - \int_t^{t+\tau} \Delta_u^i dS_u^i - \int_t^{t+\tau} r \left(C^i - S^i \frac{\partial C^i}{\partial S^i} \right) du.$$

Taking expectations gives

$$\mathbb{E}[G_{t,t+\tau}^i] = \sum_{j=1}^n \mathbb{E} \left[\int_t^{t+\tau} \lambda_v^j \frac{\partial C^i}{\partial V_f^j} du \right] + \sum_{j=1}^n \mathbb{E} \left[\int_t^{t+\tau} \lambda_\eta^j \frac{\partial C^i}{\partial \eta^j} du \right]. \quad (\text{A9})$$

9.4 Mapping theory to empirical factors

The paper uses this as a disciplined interpretation device:

- HV–IV is interpreted as a proxy for volatility risk (linked to λ_v -type premia).
- VOV is interpreted as a proxy for volatility-of-volatility risk (linked to λ_η -type premia).
- Cash is treated as a proxy for option liquidity / balance-sheet related pricing, consistent with recent evidence in the options literature.
- EWOP captures contract-market-wide variation and is empirically extremely close to the first PCA factor.

10 Short summary

1. **Delta-hedged equity option returns have a very strong factor structure.** A small number of factors explains a large portion of both time-series and cross-sectional variation.
2. **Calls and puts mostly share the same factor structure**, but each contract type also has a contract-specific component.
3. A **parsimonious tradable 4-factor model** performs nearly as well as the latent PCA benchmark:

$$\{\text{EWOP}, \text{HV–IV}, \text{Cash}, \text{VOV}\}.$$

4. The model’s robustness (OTM options; 150 characteristics) makes it a credible *benchmark* for option-return pricing and for applications that need an interpretable SDF proxy in the equity option market.

11 Conclusion

11.1 Summary

This paper concludes that **delta-hedged equity option returns exhibit a remarkably strong and parsimonious common-factor structure**. Using a look-ahead-bias-free

construction of daily delta-hedged returns (in the spirit of Duarte et al. (2024)) and forming characteristic-sorted portfolios for both calls and puts, the authors show:

- (i) **A small number of latent factors prices the cross-section extremely well.** A sparse latent (PCA-based) factor model achieves a correlation of about **0.90 or higher** between average and predicted returns, indicating that common factors explain most cross-sectional dispersion in option portfolio returns.
- (ii) **Four latent factors are sufficient for time-series and cross-sectional fit, with a clear common vs. contract-specific split.** A four-factor PCA model effectively explains both the time-series and cross-section of delta-hedged option returns. Importantly, **three** of these factors are **common to both calls and puts** and account for roughly **90% of cross-sectional variation** across the main set of portfolios, while the remaining factor is **contract-specific** and primarily matters for the **time-series dimension** (it is not central for cross-sectional pricing).
- (iii) **A tradable, characteristic-based four-factor model matches the latent benchmark.** The paper identifies an interpretable four-factor model that performs comparably to the four-PCA-factor benchmark:

$$\{\text{EWOP}, \text{HV-IV}, \text{Cash}, \text{VOV}\}.$$

Here EWOP is the equal-weighted option-portfolio return, HV-IV is the log difference between historical realized volatility and implied volatility, Cash is corporate cash holdings scaled by total assets, and VOV is volatility of implied volatility. This parsimonious tradable model reproduces the latent model’s high pricing performance and remains robust across moneyness (including out-of-the-money options) and a much larger set of portfolios (e.g., 2,990 portfolios built from 150 characteristics).

- (iv) **Option factors are distinct from stock return factors.** Traditional stock return factors have little ability to explain these option factors, implying that **higher-order moment (volatility-related) risk premia are driven by fundamentally different sources than the standard equity risk premium.**

11.2 Why these factors emerge

The economic intuition is that **delta-hedging removes (most) contemporaneous equity-directional exposure**, so the remaining option P&L reflects compensation for **volatility and higher-moment risks** and for **option-market frictions**.

1) Volatility risk premium $\Rightarrow$ HV-IV. When implied volatility is high relative to realized volatility, selling options tends to earn positive delta-hedged gains. Thus HV-IV naturally

proxies for variation in the **variance risk premium** and the compensation investors require for bearing volatility fluctuations.

2) Volatility-of-volatility risk $\Rightarrow$ VOV. Options are convex in volatility, so changes in volatility (and uncertainty about volatility itself) are first-order risks for delta-hedged returns. A VOV factor captures priced exposure to **second-order uncertainty** (volatility-of-volatility), which is central for option portfolios.

3) Limits to arbitrage / option-market liquidity $\Rightarrow$ Cash. The Cash factor (based on corporate cash-to-assets) acts as a proxy for **option liquidity / balance-sheet related pricing**, consistent with the idea that option returns embed compensation for intermediaries' and traders' constraints that vary across firms and over time.

4) An option-market aggregate component $\Rightarrow$ EWOP (contract-specific). Even after accounting for cross-sectional volatility and liquidity characteristics, there remains an option-market-wide component. EWOP captures this **aggregate option-return variation** and is empirically very close to the leading PCA factor. It is “contract-specific” in the sense that it is crucial for matching time-series comovement within calls or within puts, but it is not the main driver of cross-sectional expected-return differences once the common factors are included.

11.3 Practical implication (how to use the conclusion)

A direct implication is that the four-factor model provides a **benchmark SDF proxy for delta-neutral option strategies** (vega/gamma-exposed trades such as delta-neutral straddles, strangles, butterflies, iron condors, etc.). In empirical applications, this factor model offers a disciplined way to ask whether an option strategy earns abnormal returns or is simply compensation for systematic exposures to volatility, volatility-of-volatility, and option-market-wide/liquidity forces.

11.4 Takeaway

Delta-hedged equity option returns are largely spanned by a small set of common factors, and a tradable four-factor model (EWOP, HV-IV, Cash, VOV) matches a strong latent PCA benchmark, while remaining largely unrelated to standard stock return factors, highlighting distinct volatility and higher-moment risk premia in option markets.